MAT457 Lecture Notes

Arky!! :3c and Isaac Clark

'26 Fall Semester

Contents

1 Day 1: Here we go again! (Sep. 9, 2026) 2

2 Day 2 - σ-algebras, measures (Sep 14, 2026) 4

3 Day 3 - Outer measures 8

4 Day 4 - TA Guest Lecture! and Caratheodory's theorem 10

5 Day 5 - More on Caratheodory 13

6 Day 6 - Regularity, Cantor sets 14

7 Day 7 - Measurable functions 16

8 Day 8 - Simple functions, integration 18

9 Day 9 - Dominated Convergence theorem 21

10 Day 10 - Lusin's and Egorff's theorems 23

1 Day 1: Here we go again! (Sep. 9, 2026)

Nabutovsky's office is in BA6104. Exams will be "difficult", and you will be responsible for statements and proofs of theorems. You should know the meaning of the material. Unlike most Nabutovsky courses, tests will not be based on homework questions.

Undergraduate students are expected to know the basics of metric spaces, topological spaces, and set cardinality. You're also expected to have some exposure to the Lebesgue measure on $\mathbb{R}$ and $\mathbb{R}^n$, along with some of its properties.

Nabutovsky proceeds to survey the class about what they know. Here are the things he's asked (as to who is familiar on these).

1. Lebesgue density theorem in $\mathbb{R}^n$.
2. Fubini's theorem.
3. L^p spaces, and how to prove that they are complete.
4. Why care about the Lebesgue integral when the Riemann integral exists? (Per Nabutovsky, most functions you deal with in PDEs and other subjects are smooth, for which the Riemann integral will suffice. So why care?)

"Now I know where the class stands, and I will explain all these to you in due course." In this course, we will extend the theory of Lebesgue integration to abstract spaces. Nabutovsky believes in starting slowly (with the later part of the course being much faster). We will start with the Lebesgue measure on $\mathbb{R}$.

Consider the space $X = \mathbb{R}$. Let us define a function $f\colon \mathcal{P}(\mathbb{R}) \to [0, +\infty]$ (observe that $+\infty$ is allowed). We want to make f a *measure*; in order to do that, it must satisfy

1. $f(\emptyset) = 0$,
2. (Countable additivity). If $E_1, E_2, \dots$ is a countable collection of disjoint sets, then
$$f\left(\bigcup_i E_i\right) = \sum_i f(E_i).$$
In particular, it follows from the above that if some E_i is of infinite measure, then $\bigcup_i E_i$ is also of infinite measure.

The following two properties apply only to the case when $X = \mathbb{R}$.

3. $f([0,1]) = 1$,
4. f is translation-invariant, i.e., $f(E + t) = f(E)$, for all $E \in \mathcal{P}(\mathbb{R})$ and all $t \in \mathbb{R}$. Note that the set $E + t$ is given by $E + t = \{x \mid x - t \in E\}$.

At this point, Nabutovsky states that properties (iii) and (iv) are "kind of arbitrary". Nabutovsky also said that there is in fact no function f that satisfies all 4 of these properties.

Proof. Consider the set $\mathbb{R}/\mathbb{Q}$. For convenience, let us restrict X down to $(0,1)$; we say that $x \sim y$ if $y = x \overset{\circ}{+} q$, where $q \in \mathbb{Q}$, where the special addition symbol is defined by
$$a \overset{\circ}{+} b = \begin{cases} a + b & \text{if } a + b \in (0,1), \\ a + b - 1 & \text{if } a + b \geq 1. \end{cases}$$

Let E consist of exactly one element from each of the equivalence classes of $X/\mathbb{Q}$ (from the axiom of choice)[1], for which the cardinality of E is $\mathbb{R}$. Now, define $E_q \coloneqq E \overset{\circ}{+} q$ for $q \in \mathbb{Q} \cap [0,1)$; then, for $q_1 \neq q_2$, we see that $E_{q_1} \cap E_{q_2} = \emptyset$, since, for all $t_1, t_2 \in E$ satisfying $t_1 + q_1 = t_2 + q_2$, it must be that

$$t_1 = t_2 + (q_2 - q_1) \implies t_1 \sim t_2,$$

so $t_1 = t_2$. Now,

$$0 < \mu\big((0,1)\big) = \sum_{q \in \mathbb{Q} \cap [0,1)} \mu(E_q).$$

By translation-invariance, it must be that $\mu(E_{q_1}) = \mu(E_{q_2})$, so we have two cases to consider: either $\mu(E_q) = 0$ (for any $q \in \mathbb{Q} \cap [0,1)$), or it is greater than 0. In either case, we would get that the sum is 0 or $+\infty$, which contradicts property (iii). □

At this point, Nabutovsky gave an example of another measure.

Definition 1 (Counting measure)**.** *The counting measure counts the number of elements in a set; if there are infinitely many elements, the measure is infinite; if the set is finite, then the measure is exactly the number of elements in the set, per its definition.*

Definition 2 (Algebra)**.** *Let X be a metric (or a topological space). Let $M \subset \mathcal{P}(X)$ (with $M \neq \emptyset$). We say M is an algebra if:*

1. *If $E \in M$, then its complement E^c is in M.*
2. *If $E_1, \ldots, E_n \in M$, then their union $\bigcup_{i=1}^n$ is in M.*

Definition 3 (σ-algebra)**.** *In addition to the above properties, we ask that the union of any countable collection of elements of M is also in M.*

Using the fact that $A \cap B = (A^c \cup B^c)^c$, we also get that the countable intersection of elements of M is also in M.

[1] note: this is a Vitali set

2 Day 2 - σ-algebras, measures (Sep 14, 2026)

Last time we defined the notion of a σ-algebra. Let us begin with some examples:

1. $\mathcal{M} = \{\varnothing, X\}$.
2. $\mathcal{M} = \mathcal{P}(X)$.
3. $\mathcal{M} = \{A \subseteq X \ : \ A \text{ or } A^c \text{ is countable}\}$.[2]

Suppose that we have some collection of subsets $A_\alpha \subseteq X$ and would like to find some σ-algebra $\mathcal{M}$ so that $A_\alpha \in \mathcal{M}$ for all α. First, observe that if $\mathcal{M}_\beta$ is some collection of σ-algebras on X and $A, A_1, A_2, ... \in \mathcal{M}_\beta$ for all β, then for each β $A^c \in \mathcal{M}_\beta$ and $\bigcup_{j=1}^\infty A_j \in \mathcal{M}_\beta$, as $\mathcal{M}_\beta$ is a σ-algebra, and thus $\bigcap_\beta \mathcal{M}_\beta$ is also a σ-algebra. In particular, the intersection of all σ-algebras containing all A_α is also a σ-algebra containing all the A_α.[3] Indeed this is also the smallest σ-algebra containing all of the A_α. We call this construction the σ-algebra generated by the A_α.

If X is a metric (or topological) space, we define the Borel σ-algebra of X to be the σ-algebra generated by the open subsets. A Borel set is an element of the Borel σ-algebra. If X is, for instance, separable,[4] then the Borel σ-algebra is generated by the open balls.[5] This is not the case in general. Consider $X = \mathbb{R}$ with the discrete metric $d(x, y) = 1$ if and only if $x \neq y$ and 0 otherwise. In this case, the σ-algebra generated by all open sets is clearly $\mathcal{P}(\mathbb{R})$, but the σ-algebra generated by the open balls is simply $\{\varnothing, \mathbb{R}\}$.

Let us devote some attention to the Borel σ-algebra of $\mathbb{R}$.

Proposition 4. *The Borel σ-algebra of $\mathbb{R}$ is generated by any of the following:*

1. *The set of all open intervals (a, b).*
2. *The set of all closed intervals $[a, b]$.*
3. *The set of all half-open intervals $[a, b)$.*
4. *The set of all open rays (a, ∞).*
5. *The set of all closed rays $(-\infty, a]$.*

Proof. (2 $\subseteq$ 1) $[a, b] = \bigcap_{n=1}^\infty (a - 1/n, b + 1/n)$. (3 $\subseteq$ 2) $[a, b) = \bigcup_{n=1}^\infty [a, b - 1/n]$. (4 $\subseteq$ 3) $[a, \infty) = \bigcup_{n=1}^\infty [a, n)$ and $(a, \infty) = \bigcup_{n=1}^\infty [a + 1/n, \infty)$. (5 $\subseteq$ 4) $(-\infty, a] = (a, \infty)^c$. (1 $\subseteq$ 5) $(a, b] = (-\infty, b] \cap (-\infty, a]^c$. $(a, b) = \bigcup_{n=1}^\infty (a, b - 1/n]$. And so the σ-algebras generated by any of (1)-(5) are equal. Since the Borel σ-algebra of $\mathbb{R}$ is, by the above remark, generated by the open balls, we are done. □

Detour: what exactly are the Borel subsets of $\mathbb{R}$? Clearly the Borel σ-algebra contains all open, closed, G_δ, and F_σ sets, but it also contains all countable intersections of F_σ sets. Let us introduce

[2] It is clear that $\varnothing \in \mathcal{M}$ and that $\mathcal{M}$ is closed under complementation. Suppose $A_1, ... \in \mathcal{M}$. If all the A_j are countable then $\bigcup_{j=1}^\infty A_j$ is countable and we are done. Otherwise, A_k^c is countable for some k, but then $(\bigcup_{j=1}^\infty A_j)^c = \bigcap_{j=1}^\infty A_j^c \subseteq A_k^c$, and so is countable.

[3] This collection is non-empty as, for instance, $\mathcal{P}(X)$ is a σ-algebra containing all the A_α.

[4] According to Nabu, we will almost always assume this to be the case.

[5] It is easy to check that, in this case, every open set is the countable union of open balls.

some notation. Let us say that A is a $\mathbf{\Pi}_n$ set if it can be written of the form

$$\bigcap_{i_1=1}^{\infty}\bigcup_{i_2=1}^{\infty}\cdots\left(\bigcup/\bigcap\right)_{i_n=1}^{\infty}A_{i_1,i_2,\ldots,i_n}$$

where the $A_{i_1,i_2,\ldots,i_n}$ are either open or closed, i.e. starting[6] with an intersection then alternating between intersection and union a total of n times. Simiarly, let us say that A is a $\mathbf{\Sigma}_n$ set if it can be written of the form

$$\bigcup_{i_1=1}^{\infty}\bigcap_{i_2=1}^{\infty}\cdots\left(\bigcup/\bigcap\right)_{i_n=1}^{\infty}A_{i_1,i_2,\ldots,i_n}$$

where again the $A_{i_1,i_2,\ldots,i_n}$ are either open or closed. Is every Borel subset of $\mathbb{R}$ a $\mathbf{\Pi}_n$ or $\mathbf{\Sigma}_n$ for some n? No! Indeed, this recipe can be generalized to arbitrary countable ordinals[7] He then alluded to the continuum hypothesis. My take on the point we was trying to make is how we may use transfinite induction to "explicitly" determine all of the Borel subsets by iterating the above process over the countable ordinals, a procedure which "terminates at the first uncountable ordinal."[8] Nabu also made the observation that the cardinality of the Borel σ-algebra of $\mathbb{R}$ is the cardinality of the continuum, so that's something.

Later, we will often prefer to work with disjoint unions as opposed to arbitrary unions. This is where "disjointification" comes in. Given $A_n \in \mathcal{M}$, we may define $B_1 = A_1$ and $B_{n+1} = A_{n+1} \cap \bigcap_{j=1}^{n} A_j^c$. Note that the B_m are disjoint, lie in $\mathcal{M}$, and $\bigcup_{n=1}^{\infty} A_n = \bigcup_{m=1}^{\infty} B_m$.

Suppose we have some collection of measurable spaces[9] $(X_\alpha, \mathcal{M}_\alpha)$ and want to define a σ-algebra on the product $\prod_\alpha X_\alpha$, which we'll denote by[10] $\bigotimes_\alpha \mathcal{M}_\alpha$. To do this, we mimick the construction of the product topology in topology, and take the σ-algebra generated by the preimages $\pi_\beta^{-1}(A)$ for $A \in \mathcal{M}_\beta$,where $\pi_\beta : \prod_\alpha X_\alpha \to X_\beta$ is the projection map. For example, consider $\mathbb{R}\times\mathbb{R}$ where we take each copy of $\mathbb{R}$ to have the Borel σ-algebra. Since the open intervals generate the Borel σ-algebra of $\mathbb{R}$, it follows[11] that the product σ-algebra is generated by the open rectangles $(a,b)\times(c,d)$. These in turn are the open balls of $\mathbb{R}$ under the metric $d_{max}(x,y) = ||x-y||_\infty = \max(|x_1-y_1|, |x_2-y_2|)$; since all norms on $\mathbb{R}^2$ are equivalent, the product σ-algebra is also generated by the usual open balls of $\mathbb{R}^2$, i.e. the product σ-algebra of $\mathbb{R}\times\mathbb{R}$ and the Borel σ-algebra of $\mathbb{R}^2$ coincide.

Definition 5 (Measure)**.** *Let $(X, \mathcal{M})$ be a measurable space. A measure is a map $\mu : \mathcal{M} \to [0,\infty]$ such that:*

1. *$\mu(\varnothing) = 0$.*
2. *"Countable additivity" If $A_n \in \mathcal{M}$ are disjoint then $\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n)$.*

Let's see some examples:

1. $\mathcal{M} = \mathcal{P}(X)$ and $\mu(A) = \infty$ if A has infinitely many elements and $\mu(A) = |A|$ (the cardinality of A) otherwise. More generally, given $f : X \to [0,\infty]$ we may define $\mu_f(A) = \sum_{a\in A} f(a)$.
2. $\mathcal{M} = \mathcal{P}(X)$ and $\mu(A) = 0$ if A is countable and ∞ otherwise.

[6] Reading left-to-right.

[7] An ordinal is an equivalence class of ordered sets.

[8] Someone once told me that transfinite induction was first used/formulated to "describe" the Borel subsets of $\mathbb{R}$; so be sure to memorize that for pub trivia.

[9] A measurable space is a non-empty set X equipped with a σ-algebra $\mathcal{M}$.

[10] :(.

[11] Exercise!

3. Given a measure $\mu_0 : \mathcal{M}_0 \to [0,\infty]$ and $A \subseteq X$, define $\mathcal{M} = \{B \subseteq X \ : \ A \cap B \in \mathcal{M}_0\}$ and $\mu : \mathcal{M} \to [0,\infty]$ by $\mu(B) = \mu_0(A \cap B)$. "The restriction measure"
4. Given $x_0 \in X$, define $\mu(A) = 1$ if $x_0 \in A$ and 0 otherwise. "Dirac measure"
5. Lebesgue measure on $\mathbb{R}^n$.

The following is a non-example. Define $\mu(A) = \infty$ if A is infinite and 0 otherwise.

Definition 6 (Finite)**.** *A measure μ is finite if $\mu(X) < \infty$.*

For example, $\mu(A) = \int_A e^{-x^2}\,dx$ on the σ-algebra of Lebesgue measurable subsets of $\mathbb{R}$, the Dirac measure. A non-example: the Lebesgue measure.

Definition 7 (σ-finite)**.** *A measure μ is σ-finite if there are countably many measurable sets E_i so that $\mu(E_i) < \infty$ and $X = \bigcup_{i=1}^{\infty} E_i$.*

For example, the Lebesgue measure. A non-example: the counting measure on $\mathbb{R}$.

Definition 8 (Semi-finite)**.** *A measure μ is semi-finite if $E \in \mathcal{M}$ and $\mu(E) = \infty$ implies there is some $F \subseteq E$ with $F \in \mathcal{M}$ and $0 < \mu(F) < \infty$.*

Exercise 9. *Show that σ-finite implies semi-finite, but that the converse may fail.*

Definition 10 (Measure space)**.** *A measure space is a triple $(X, \mathcal{M}, \mu)$ where X is a non-empty set, $\mathcal{M}$ is a σ-algebra on X, and $\mu : \mathcal{M} \to [0,\infty]$ is a measure.*

Theorem 11. *Let $(X, \mathcal{M}, \mu)$ be a measure space. We have:*

1. *"Monotonicity" If $A, B \in \mathcal{M}$ and $A \subseteq B$ then $\mu(A) \leq \mu(B)$.*
2. *"Countable subadditivity" If $A_n \in \mathcal{M}$ then $\mu(\bigcup_{n=1}^{\infty} A_n) \leq \sum_{n=1}^{\infty} \mu(A_n)$.*
3. *"Continuity from below" If $E_1 \subseteq E_2 \subseteq \cdots$ then $\mu(\bigcup_{n=1}^{\infty} E_n) = \lim_{n\to\infty} \mu(E_n)$.*
4. *"Continuity from above" If $E_1 \supseteq E_2 \supseteq \cdots$, $\mu(E_1) < \infty$ then $\mu(\bigcap_{n=1}^{\infty} E_n) = \lim_{n\to\infty} \mu(E_n)$.*

Proof. For (1), note that B may be written as the disjoint union of A and $B \cap A^c$, so by additivity $\mu(B) = \mu(A) + \mu(B \cap A^c) \geq \mu(A)$, since μ is non-negative.[12] For (2), apply disjointification, countable additivity, and monotonicity to obtain $\mu(\bigcup_{n=1}^{\infty} A_n) = \mu(\bigcup_{m=1}^{\infty} B_m) = \sum_{m=1}^{\infty} \mu(B_m) = \sum_{m=1}^{\infty} \mu(A_m \cap \bigcap_{j=1}^{m-1} A_j) \leq \sum_{m=1}^{\infty} \mu(A_m)$. For (3), take $B_1 = E_1$ and $B_{m+1} = E_{m+1} \setminus E_m$. Since the B_m are disjoint and $\bigcup_{n=1}^{\infty} E_n = \bigcup_{m=1}^{\infty} B_m$, we have $\mu(\bigcup_{n=1}^{\infty} E_n) = \mu(\bigcup_{m=1}^{\infty} B_m) = \sum_{m=1}^{N} \mu(B_m) + \sum_{m=N+1}^{\infty} \mu(B_m) = \mu(\bigcup_{m=1}^{N} B_m) + \sum_{m=N+1}^{\infty} \mu(B_m) = \mu(E_N) + \sum_{m=N+1}^{\infty} \mu(B_m)$, where the latter summand is the tail of a convergent series, and so vanishes in the limit. For (4), put $F_j = F_1 \setminus E_j$. Notice that $F_1 \subseteq F_2 \subseteq \cdots$ and so by (3) we have $\mu(\bigcup_{j=1}^{\infty} F_j) = \lim_{j\to\infty} \mu(F_j)$. Now, $\mu(F_j) = \mu(E_1) - \mu(E_j)$ by additivity; and $\bigcup_{j=1}^{\infty} F_j = E_1 \setminus \bigcap_{n=1}^{\infty} E_n$. All together, we have $\mu(E_1) - \mu(\bigcap_{n=1}^{\infty} E_n) = \mu(\bigcup_{j=1}^{\infty} F_j) = \mu(E_1) - \lim_{n\to\infty} \mu(E_n)$. Since $\mu(E_1) < \infty$, we conclude. □

Note that the hypothesis in (4) that $\mu(E_1) < \infty$ is important. Consider the Lebesgue measure on $\mathbb{R}$ and $E_i = (i, \infty)$. Then $\bigcap_{i=1}^{\infty} E_i = \varnothing$ but $\lim_{i\to\infty} \mu(E_i) = \infty$.

A measurable set of measure zero is called a null set. By monotonicity, measurable subsets of null sets are null sets. However, it could be the case that a subset of a null set fails to be measurable. For a trivial example, take $X = \{a, b\}$, $\mathcal{M} = \{\varnothing, X\}$, and μ the counting measure.

[12] Later in the course, we will discuss signed measures, which are precisely what it says on the tin.

Definition 12 (Complete measure)**.** *A measure is said to be complete*[13] *if all subsets of null sets are measurable.*

Definition 13 (Completion)**.** *Let $(X, \mathcal{M}, \mu)$ be a measure space. Let $\overline{\mathcal{M}}$ be the σ-algebra generated by $\mathcal{M}$ and the subsets of null sets. The (unique) extension of μ to $\overline{\mathcal{M}}$ is called the completion.*

Nabu claimed that $\overline{\mathcal{M}} = \{A \cup N \ : \ A \in \mathcal{M}, N \subseteq Z \in \mathcal{M} \text{ with } \mu(Z) = 0\}$. He didn't have time for a proof, but I've provided one below. While technically it should be a theorem that the extension of a measure actually exists, this result makes it clear that setting $\overline{\mu}(A \cup N) = \mu(A)$ is well-defined. Uniqueness, along with some other checks, are straight-forward.[14]

Proposition 14. *$\overline{\mathcal{M}} = \{A \cup N \ : \ A \in \mathcal{M}, N \subseteq Z \in \mathcal{M}$ with $\mu(Z) = 0\}$.*

Proof. $\varnothing$ is a null set, so $\varnothing = \varnothing \cup \varnothing \in \overline{\mathcal{M}}$. By countable subadditivity, the countable union of null sets is a null set, taken with the identity $\bigcup_{n=1}^{\infty} A_n \cup N_n = (\bigcup_{n=1}^{\infty} A_n) \cup (\bigcup_{n=1}^{\infty} N_n)$, it quickly follows that $\overline{\mathcal{M}}$ is closed under countable unions. Now suppose $E = A \cup N \in \overline{\mathcal{M}}$ and pick $Z \in \mathcal{M}$ with $N \subseteq Z$ and $\mu(Z) = 0$. Now, $N \subseteq Z$ implies $Z^c \subseteq N^c$ and so $N^c = Z^c \cup (Z \setminus N)$. Substituting this in, we have $E^c = A^c \cap N^c = (A^c \cap Z^c) \cup (A^c \cap (Z \setminus N))$. But $A^c, Z^c \in \mathcal{M}$ and $A^c \cap (Z \setminus N) \subseteq Z$ which has measure zero, and thus E^c is of the desired form. □

[13] This terminology has nothing to do with the notion of completeness for metric spaces.
[14] Exercise!

3 Day 3 - Outer measures

Last time: we talked about complete measures and completions of measures. We started with the proposition from last time about the σ-algebra generated by $\mathcal{M}$ and the subsets of null sets being all those sets of the form $M \cup N$ for $M \in \mathcal{M}$ and $N \subseteq Z$ for some $Z \in \mathcal{M}$ with $\mu(Z)$. The proof Nabu gave in lecture is essentially the one I gave in Day 2. Nabu also spelled out why the extension $\overline{\mu}(M \cup N) = \mu(M)$ is well-defined. Suppose that $M_1 \cup N_1 = M_2 \cup N_2$ for M_i, N_i, and $N_i \subseteq Z_i$ as before; we need to show that $\mu(M_1) = \mu(M_2)$. By symmetry, it suffices to show that $\mu(M_1) \leq \mu(M_2)$. Now, $M_1 \subseteq M_1 \cup N_1 = M_2 \cup N_2 \subseteq M_2 \cup Z_2$ and so by monotonicity and subadditivity, $\mu(M_1) \leq \mu(M_2) + \mu(Z_2) = \mu(M_2)$, and we are done.

We now move our attention to so-called "outer measures."

Definition 15 (Outer measure)**.** *A function $\mu^* : \mathcal{P}(X) \to [0, \infty]$ is an outer measure if:*[15]

1. *$\mu^*(\varnothing) = 0$.*
2. *μ^* has monotonicity.*
3. *μ^* is countably subadditive.*

Differences: "no pesky σ-algebras;" measures have monotonicity and countable subadditivity, but also countable additivity (on disjoint sets). Question: "What's so outer about this measure?"

To answer this, let's consider an example on $X = \mathbb{R}$. Let us start by setting the outer measure of an interval[16] to be its length. Then define $\mu^*(A) = \inf\{\sum_{i=1}^{\infty}(b_i - a_i) \ : \ \bigcup_{i=1}^{\infty}(a_i, b_i) \supseteq A\}$.[17] Recall that this is the first step in defining the Lebesgue measure.

Question: do we really need countable collections? Yes, since there is no finite collection of intervals which cover $\mathbb{R}$. What if we restrict to bounded sets? The set $\mathbb{Q} \cap [0, 1]$ is bounded and has outer measure 0, as it should, but if we restrict to finite collections, then the sums of lengths of intervals of any finite covering of $\mathbb{Q} \cap [0, 1]$ is at least 1,[18] and so this notion of outer measure would not play well with limits, which messes with the completeness of spaces of integrable functions.

We can mimick this construction in much more generality.

Theorem 16. *Suppose that $\mathcal{E} \subseteq \mathcal{P}(X)$ such that $\varnothing, X \in \mathcal{E}$ and that $\lambda : \mathcal{E} \to [0, \infty]$ satsifies $\lambda(\varnothing) = 0$. Define $\mu^*(A) = \inf\{\sum_{i=1}^{\infty}\lambda(E_i) \ : \ \bigcup_{i=1}^{\infty} E_i \supseteq A, \ E_i \in \mathcal{E}\}$. Then, μ^* is an outer measure.*

Proof. Note that μ^* is well-defined since $\{X\} \subseteq \mathcal{E}$ is a countable cover of any subset $A \subseteq X$. If $A \subseteq B$ then every covering of B is also a covering of A, and so $\mu^*(A) \leq \mu^*(B)$ since the infimum over a larger set is at most the infimum over a smaller set. Now, fix $\varepsilon > 0$ and suppose $A_i \subseteq X$. Fix i, by definition, we may pick $E_{i,j} \in \mathcal{E}$ so that $\bigcup_{j=1}^{\infty} E_{i,j} \supseteq A_i$ and $\sum_{j=1}^{\infty}\lambda(E_{i,j}) \leq \mu^*(A_i) + 2^{-i}\varepsilon$. Then $\{E_{i,j}\}$ is a countable covering of $\bigcup_{i=1}^{\infty} A_i$ by sets in $\mathcal{E}$, and so $\mu^*\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty}\sum_{j=1}^{\infty}\lambda(E_{i,j}) \leq \sum_{i=1}^{\infty}\mu^*(A_i) + 2^{-i}\varepsilon = \varepsilon + \sum_{i=1}^{\infty}\mu^*(A_i)$. Since ε was arbitrary, we are done. □

[15] The notions of monotonicity and countable subadditivity coincide with their use in **Theorem 11**

[16] Of any kind.

[17] In this example we've required the covering to consist of open intervals, but nothing changes by replacing these with some other interval.

[18] Convince yourself, using topology, that if finitely many open intervals cover $\mathbb{Q} \cap [0, 1]$ then they must also cover $[0, 1]$.

So it is very easy to construct outer measures. Next, much like how the above theorem generalizes the outer measure of $\mathbb{R}$, we can also mimick the construction of the Lebesgue measure from the outer measure at the level of generality of the above.

Definition 17 (μ^*-measurable)**.** *Given an outer measure μ^*, we say a set $A \subseteq X$ is μ^*-measurable if for all $E \subseteq X$, $\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c)$.*

And, as in the case of the Lebesgue measure, this procedure produces a (complete) measure.

Theorem 18 (Caratheodory)**.** *If μ^* is an outer measure on X, then*

1. *The collection of μ^*-measurable sets, which we'll denote by $\mathcal{M}$, is a σ-algebra.*
2. *The restriction of μ^* to $\mathcal{M}$ is a complete measure.*

We shall delay the proof for now. Let us discuss what we don't get from Caratheodory's theorem. We don't get any real description of $\mathcal{M}$ in this theorem, a priori it could be that $\mathcal{M} = \{\varnothing, X\}$. We are also not guaranteed that the restriction of μ^* to $\mathcal{E} \cap \mathcal{M}$ should agree with λ. "In such generality we don't often get what we want."

Let us argue why the restriction of the outer measure to intervals does agree with length. Without loss of generality, we may assume the coverings in the definition of the outer measure consist of open intervals. Similarly, it suffices to show the claim for closed intervals.[19] Fix a countable cover $I_i = (a_i, b_i)$ of $[a, b]$. By compactness, there exists K so that $[a, b] \subseteq \bigcup_{i=1}^{K} I_i$. Fix a large number N and consider the scaled intervals (Na_i, Nb_i) and $[Na, Nb]$. Let S be the number of integral lattice points in $[Na, Nb]$ and let S_i be the number of integral lattice points in (Na_i, Nb_i). By the standard bounds for the floor and ceiling functions, it follows that $S_i \leq N(b_i - a_i) + 1$ and $S \geq N(b-a) - 1$. Since $I_1, ..., I_K$ cover $[a, b]$, we have $S \leq \sum_{i=1}^{K} S_i$. All together,

$$b - a - \frac{1}{N} \leq \frac{S}{N} \leq \sum_{i=1}^{K} \frac{S_i}{N} \leq \frac{K}{N} + \sum_{i=1}^{K} (b_i - a_i) \leq \frac{K}{N} + \sum_{i=1}^{\infty} (b_i - a_i)$$

Taking N to infinity, we obtain $b - a \leq \sum_{i=1}^{\infty} (b_i - a_i)$ and so $b - a \leq \mu^*([a, b])$. The other inequality is clear since $\{[a, b]\}$ is a countable cover of $[a, b]$ with total length $b - a$. □

[19] Due to time reasons, Nabu couldn't finish the proof of this claim. What follows this footnote is my fleshing out of what he vibe-coded, and therefore could be completely different from what he had in mind.

4 Day 4 - TA Guest Lecture! and Caratheodory's theorem

Today's lecture was given our TA.[20] Most of the lecture will be guided by the question: "is there a Borel measure?" Of course, we will answer this in much more generality.

Definition 19 (Premeasure)**.** *Let $\mathcal{A}$ be an algebra.*[21] *A premeasure is a map $\mu_0 : \mathcal{A} \to [0, \infty]$ satisfying:*

1. *$\mu_0(\varnothing) = 0$.*
2. *Countable additivity (for disjoint unions).*

Roughly speaking, we will follow these steps: (1) define a premeasure on $\mathcal{A}$,[22] (2) get an outer measure from this premeasure, (3) get a σ-algebra $\mathcal{M}$ from this outer measure so that the restriction to $\mathcal{M}$ is a complete measure, and (4) show that this complete measure is "almost" unique.

Let $\mathcal{A}$ be the smallest algebra containing all intervals of the form $(a, b]$, $a < b$. Note that since $\mathcal{A}$ is only an algebra, it is distinct from, say, the smallest algebra containing all open intervals.

Proposition 20. *$E \in \mathcal{A}$ if and only if $E = \bigcup_{j=1}^{n} (a_j, b_j]$.*

Proof. It suffices to show that the sets of this form are an algebra. Taking $n = 0$ yields $\varnothing$. Closure under finite unions is clear. For complementation, observe that $(a_j, b_j]^c = (-\infty, a_j] \cup (b_j, \infty)$, and that for a general union we can "fill in the gaps" using intervals of the correct form. □

From **Proposition 20**, it is clear that defining $\mu_0(E) = \sum_{j=1}^{n} b_j - a_j$ gives a premeasure. This completes step 1 for the construction of the Borel measure.

Proposition 21. *Let $\mathcal{A}$ be an algebra and μ_0 a premeasure on $\mathcal{A}$. Define, as before, for $E \subseteq X$, $\mu^*(E) = \inf\{\sum_{j=1}^{\infty} \mu(A_j) \ : \ A_j \in \mathcal{A}, \ E \subseteq \bigcup_{j=1}^{\infty} A_j\}$. Then, μ^* is an outer measure. Further, μ^* extends μ_0.*

Proof. That μ^* is an outer measure was seen before as **Theorem 16**. Since $\{E\}$ is always a countable cover of E, it is clear that $\mu_0(E) \geq \mu^*(E)$. For the other inequality, notice it suffices to show that $\mu_0(E) \leq \sum_{j=1}^{\infty} \mu_0(A_j)$ for any countable cover $A_j \in \mathcal{A}$ of E. But this follows from monotonicity and then countable subadditivity.[23] □

The above proof should be contrasted with the one Nabu tried to give for the Lebesgue measure. In particular, starting with a premeasure on $\mathcal{A}$ instead of only having λ of intervals helps a lot. This wraps up step 2.

The jump from an outer measure to a suitable σ-algebra is called Caratheodory's theorem. Recall that we say $A \subseteq X$ is μ^*-measurable if for all $E \subseteq X$ we have $\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c)$.

Proof of Caratheodory's theorem (**Theorem 18**). Let $\mathcal{M}$ be all those sets which are μ^*-measurable. On Day 3 we showed that $\varnothing \in \mathcal{M}$, $\mathcal{M}$ is closed under complementation, and that $\mu^*(\varnothing) = 0$. Let

[20] Or, perhaps, Nabu after very extensive plastic surgery.

[21] Like a σ-algebra, except we only require closure under finite unions.

[22] This is the only place we will use the specifics that lead to the Borel measure.

[23] Essentially the same argument that a measure is countably subadditive also demonstrates that a premeasure is countably subadditive.

us show that $\mathcal{M}$ is closed under finite unions and that $\mu^*|_{\mathcal{M}}$ is finitely additive. For $A, B \in \mathcal{M}$, $E \subseteq X$, we have:

$$\begin{aligned}\mu^*(E) &= \mu^*(E \cap A) + \mu^*(E \cap A^c) \\ &= \mu^*((E \cap A) \cap B) + \mu^*((E \cap A) \cap B^c) + \mu^*((E \cap A^c) \cap B) + \mu^*((E \cap A^c) \cap B^c) \\ &\geq \mu^*(E \cap (A \cup B)) + \mu^*(E \cap (A \cup B)^c) \\ &\geq \mu^*(E)\end{aligned}$$

And so $A \cup B \in \mathcal{M}$. Applying this to $E = A$ in the case where A, B are disjoint, we have $\mu^*(A \cup B) = \mu^*((A \cup B) \cap A) + \mu^*((A \cup B) \cap A^c) = \mu^*(A) + \mu^*(B)$. And so we have finite additivity. Let us now show that $\mathcal{M}$ is closed under countable unions and that $\mu^*|_{\mathcal{M}}$ is countably additive. By disjointification, we may assume without loss of generality that the $A_j \in \mathcal{M}$ are disjoint. Define $B_n = \bigcup_{j=1}^n A_j$, $B = \bigcup_{j=1}^\infty A_j$. Let $E \subseteq X$. He left as an exercise that $\mu^*(E \cap B_n) = \sum_{j=1}^n \mu^*(E \cap A_n)$. By the finite case, $B_n \in \mathcal{M}$ and so $\mu^*(E) = \mu^*(E \cap B_n) + \mu^*(E \cap B_n^c) \geq \sum_{j=1}^n \mu^*(E \cap A_j) + \mu^*(E \cap B^c)$. Taking $n \to \infty$, we have

$$\begin{aligned}\mu^*(E) &\geq \sum_{j=1}^\infty \mu^*(E \cap A_j) + \mu^*(E \cap B^c) \\ &\geq \mu^*\left(E \cap \bigcup_{j=1}^\infty A_j\right) + \mu^*(E \cap B^c) \\ &\geq \mu^*(E)\end{aligned}$$

And so $B = \bigcup_{j=1}^\infty A_j \in \mathcal{M}$. If the A_j are disjoint, the trick from before works to show countable additivity. It remains to show that μ^* is complete. It suffices to show that $\mu^*(A) = 0$ implies $A \in \mathcal{M}$. But this is clear from subadditivity and monotonicity: $\mu^*(E) \leq \mu^*(E \cap A) + \mu^*(E \cap A^c) \leq \mu^*(A) + \mu^*(E) = \mu^*(E)$. □

For the full step 3, we still need to show that $\mathcal{A} \subseteq \mathcal{M}$.

Proposition 22. *If $A \in \mathcal{A}$ then $A \in \mathcal{M}$.*

Proof. Let $\varepsilon > 0$, $E \subseteq X$. Pick $B_j \in \mathcal{A}$ so that $E \subseteq \bigcup_{j=1}^\infty B_j$ and $\sum_{j=1}^\infty \mu_0(B_j) \leq \mu^*(E) + \varepsilon$.[24] Then,

$$\begin{aligned}\mu^*(E) + \varepsilon &\geq \sum_{j=1}^\infty \mu_0(B_j) \\ &= \sum_{j=1}^\infty \mu_0(B_j \cap A) + \sum_{j=1}^\infty \mu_0(B_j \cap A^c) \text{ , note}^{25} \\ &= \sum_{j=1}^\infty \mu^*(B_j \cap A) + \sum_{j=1}^\infty \mu^*(B_j \cap A^c) \\ &\geq \mu^*(E \cap A) + \mu(E \cap A^c)\end{aligned}$$

and so taking $\varepsilon \to 0$ finishes the proof. □

[24] These exist by the definition of the infimum.

Let us now characterize what we mean by "almost" unique in step 4.

Proposition 23. *Let ν be another measure on $\mathcal{M}$ such that $\nu|_{\mathcal{M}} = \mu_0$. Then,*

1. *$\nu(E) \leq \mu(E)$ for all $E \in \mathcal{M}$. [here $\mu = \mu^*|_{\mathcal{M}}$]*
2. *If $\mu(E) < \infty$ then $\nu(E) = \mu(E)$.*

In particular, if μ is σ-finite then we have uniqueness.

Proof. For (1), it suffices to show that for $E \subseteq \bigcup_{j=1}^{\infty} A_j$ with $A_j \in \mathcal{A} \subseteq \mathcal{M}$, that we have the inequality $\nu(E) \leq \sum_{j=1}^{\infty} \mu_0(A_j)$. We check: $\nu(E) = \nu(\bigcup_{j=1}^{\infty} E \cap A_j) \leq \sum_{j=1}^{\infty} \nu(E \cap A_j) \leq \sum_{j=1}^{\infty} \nu(A_j) = \sum_{j=1}^{\infty} \mu_0(A_j)$. For (2), suppose $\mu(E) < \infty$ and let $\varepsilon > 0$. Pick $A_j \in \mathcal{A}$ (without loss of generality, assume they are disjoint) so that $E \subseteq \bigcup_{j=1}^{\infty} A_j$ and $\sum_{j=1}^{\infty} \mu_0(A_j) \leq \mu(E) + \varepsilon$. Let $A = \bigcup_{j=1}^{\infty} \in \mathcal{M}$. Note that $\mu(A) = \nu(A)$. Since $\mu(E) < \infty$, we also have $\mu(A \setminus E) \leq \varepsilon$. Then,

$$\mu(E) \leq \mu(A) = \nu(A) = \nu(E) + \nu(A \setminus E) \leq \nu(E) + \varepsilon$$

So taking $\varepsilon \to 0$ finishes the proof. □

Now, recall that we defined μ_0 on $E = \bigcup_{j=1}^{n} (a_j, b_j]$ to be $\sum_{j=1}^{n} b_j - a_j$. But what actually was special about the length that allowed us to define a premeasure? Nothing! Indeed,

Proposition 24 (1.15 in Folland)**.** *If $F : \mathbb{R} \to \mathbb{R}$ is increasing and right continuous, then $\mu_0 \left(\bigcup_{j=1}^{n} (a_j, b_j] \right) = F(b_j) - F(a_j)$ is a premeasure on $\mathcal{A}$.*

We can then repeat the construction we did for the case $F(x) = x$ to obtain a measure, which we'll denote by μ_F, on $\mathcal{B}(\mathbb{R})$. We then have the following propositions about this construction.

Proposition 25 (1.16, Line 2 in Folland)**.** *If F, G are increasing and right continuous then $\mu_F = \mu_G$ if and only if $F - G$ is constant.*

So this extension is essentially unique.

Proposition 26 (1.16, Line 3 in Folland)**.** *Suppose μ is a measure on some $\mathcal{M}$ such that $\mathcal{B}(\mathbb{R}) \subseteq \mathcal{M}$ with $\mu(E) < \infty$ on bounded Borel subsets $E \subseteq \mathbb{R}$. Define $F : \mathbb{R} \to \mathbb{R}$ by*

$$F(x) = \begin{cases} \mu((0, x]) & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -\mu((-x, 0]) & \text{if } x < 0 \end{cases}$$

Then F is increasing and right continuous and $\mu = \mu_F$.

Remark 27. *This construction is very similar to that of a cumulative distribution function from probability theory.*

Definition 28. *The completion of μ_F is said to be Lebesgue-Stieltjes.*

5 Day 5 - More on Caratheodory

Brief recap: Caratheodory's theorem and, in particular, the construction of the Lebesgue measure on $\mathbb{R}$. Fix $\mu : \mathcal{E} \to [0, \infty]$ where $\varnothing, X \in \mathcal{E}$ such that: $\mu(\varnothing) = 0$ and μ satisfies countable additivity. We then defined how to use this to construct an outer measure μ^*, i.e. a map $\mu^* : \mathcal{P}(X) \to [0, \infty]$ satsifies countably subadditivity and monotonicity. The idea of Caratheodory is then to restrict to those sets E with the property that $\mu*(A) = \mu^*(A \cap E) + \mu^*(A \cap E^c)$. Remarkably, these sets form a σ-algebra and the restriction of μ^* to said σ-algebra is a complete measure. Note that Caratheodory's theorem[26] only requires μ^* to be an outer measure, it need not be that μ^* is induced by some premeasure.

In practice. Let $f(A) = 1$ if $A \neq \varnothing$ and 0 if $A = \varnothing$. Let's trace through the steps to see what measure we get from this. This function is not a premeasure. This is more in line with what Nabu did with λ in the case of the Lebesgue measure. Recall that this way of doing it was easier to define out of the gate, or rather there was less to check out of the gate, but proving that the outer measure extended λ required an argument. Let $\mu^*(A) = \inf\{\sum_{i=1}^{\infty} f(E_i) \ : \ ...\}$. Let us check that this is an outer measure. Monotonicity is baked in. It is also clear that $\mu^*(\varnothing) = 0$ and that μ^* is countably subadditive. As a result, we can apply Caratheodory to this (we can't get that the resulting measure is an extension of f, as f is terrible). For simplicity, let's suppose $X = \mathbb{R}$. What are the μ^*-measurable sets? Suppose $E \neq \varnothing, \mathbb{R}$, then $1 = \mu^*(\mathbb{R}) \neq \mu^*(E) + \mu^*(E^c)$ since E, E^c are non-empty. Thus, the σ-algebra of μ^*-measurable sets is trivial. So this is disappointing on two fronts: the restriction of μ^* not an extension and no interesting measurable sets. This should be contrasted with the case from Day 4 that f was a premeasure, where we do get an extension of f and we do have that every element in the domain algebra of f is measurable.

He then recapped that, if we have an increasing and right continuous function $F : \mathbb{R} \to \mathbb{R}$, we can define a premeasure $\mu_0(\bigcup_{j=1}^{n}(a_j, b_j]) = \sum_{j=1}^{n} F(b_j) - F(a_j)$ on the algebra generated by the intervals of the form $(a, b]$.[27] And that we (essentially) have uniqueness.

Let's "remember the other story" of constructing the Lebesgue measue. We've shown that the outer measure defined in terms of λ agrees with λ on compact intervals.[28] We can generalize this argument to Lebesgue-Stieltjes by counting integral lattice points in $(F(a_j), F(b_j))$. Notice that this argument very much uses compactness, as the argument relies on a finite cover. In the argument on Day 3, I stated that we could assume without loss of generality that the coverings are by open intervals, the reason for this is that we can replace $(a_j, b_j]$ with (a_j, b_j) and $(b_j - 2^{-j}\varepsilon, b_j + 2^{-j}\varepsilon)$ without changing the fact that the cover is countable or the sum of the lengths modulo ε. We have not shown, however, that these sets are measurable![29] Let $\varepsilon > 0$. Fix $A \subseteq \mathbb{R}$. Let $A_1 = A \cap (a, \infty)$ and let $A_2 = A \cap (-\infty, a]$. Pick a cover I_j of A by intervals so that $\sum_{j=1}^{\infty} \lambda(I_j) \leq \mu^*(A) + \varepsilon$. Let $J_j = I_j \cap (a, \infty)$ and $J'_j = I_j \cap (-\infty, a]$. Note that $\lambda(I_j) = \lambda(J_j) + \lambda(J'_j)$. Then the J_j (J'_j resp.) are a countable cover of A_1 (A_2 resp.) by intervals. So $\mu^*(A_1) + \mu^*(A_2) \leq \sum_{j=1}^{\infty} \lambda(J_j) + \sum_{j=1}^{\infty} \lambda(J'_j) = \sum_{j=1}^{\infty} \lambda(I_j) \leq \mu^*(A) + \varepsilon$. This yields the claim, as the open rays generate the Borel σ-algebra.

Next question (to think about at home): what happens when we do this slash how can we do this to an arbitrary metric space?

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[26] He's referring to the formulation of Caratheodory's theorem from Day 3. As we will see, we lose pieces of the formulation we proved on Day 4 when we have less information.

[27] "Not very instructive, but nonetheless you will have to study this." - Nabu

[28] In lecture, he retold the argument, as he technically left it unfinished on Day 3. But the argument he gave is essentailly what I filled in on Day 3, so see that for details.

[29] We had one minute left at this point, so the following argument is again my fleshing out of his vibe-coding.

6 Day 6 - Regularity, Cantor sets

Last time: we defined a very general way to get σ-algebras from outer measures. In particular, we defined the Lebesgue-Stiltjes measures. The same technique can be used to construct the Lebesgue measure in $\mathbb{R}^n$. All these measures happen to agree that Borel sets are measurable. But this is a lot of machinery to define something we saw in MAT357. Question: can we do this in arbitrary metric spaces? The answer is not as easy as one may expect. Recall from last time that there were pitfalls in applying the recipe starting with only length instead of a premeasure. In $\mathbb{R}$, there is a very natural algebra to define length on, i.e. the h-intervals, and in $\mathbb{R}^n$, we can argue similarly but with the infinity norm. He then said something about recovering the lengths of curves in the plane from a suitable measure on $\mathbb{R}^2$?[30] Apparently, the punchline was that curves would not be μ^*-measurable if we set $\lambda(\text{disk of radius } r) = r$. So there's trouble ahead.

To better approach this problem, let us first consider what we need measures for.

"Regularity of Lebesgue-Stiltjes measure."

1. Let A be a (Lebesgue) measurable set. Then $\mu(A) = \inf\{\mu(U) \ : \ A \subseteq U, \ U \text{ open}\}$.
2. Let A be a (Lebesgue) measurable set. Then $\mu(A) = \sup\{\mu(C) \ : \ C \subseteq A, \ C \text{ closed}\}$.
3. Let A be a (Lebesgue) measurable set. Then $\mu(A) = \sup\{\mu(K) \ : \ K \subseteq A, \ K \text{ compact}\}$.

Definition 29 (Outer regularity)**.** *The first item is called "outer regularity."*

Definition 30 (Inner regularity)**.** *The second item is called "inner regularity."*

Definition 31 (Inner regularity with respect to compact sets)**.** *The third item is called "inner regularity with respect to compact sets."*

For outer regularity, it is enough to show that for all measurable A with $\mu(A) < \infty$ and $n \in \mathbb{N}$ there is an open set $A \subseteq U_n$ with $\mu(U_n) \leq \mu(A) + 1/n$. This can be seen to be the case for the Lebesgue measure by replacing, without loss of generality, all intervals in a covering with open intervals.

Next, consider $G = \bigcap_{n=1}^{\infty} U_n$ and let $U_1' = U_1$, $U_2' = U_1 \cap U_2$, and so on. Then $\mu(A) = \lim_{n\to\infty} \mu(U_n) = \mu(G)$, by continuity and that $\mu(U_n) \leq \mu(A) + 1/n$. And so there is a G_δ set, G, so that $A \subseteq G$ and $\mu(A) = \mu(G)$.

If A is bounded, then $\mu(A) < \infty$. Thus, if $\mu(A) = \infty$ then A is unbounded. In this case, write $A = \bigcup_{j\in\mathbb{Z}} A \cap (j, j+1]$. For each $j \in \mathbb{Z}$, $A_j = A \cap (j, j+1]$ is bounded and so has finite measure and so we can pick U_j open so that $\mu(U_j \setminus A_j) < 2^{-|j|}\varepsilon$. Then $\mu(\bigcup_{j\in\mathbb{Z}} U_j \setminus A_j) = \sum_{j\in\mathbb{Z}} \mu(U_j \setminus A_j) \leq \varepsilon$. So we can also "do this approximation" for sets with infinite measure.

Let's move on to inner regularity. Recall that compact is the same as closed and bounded by Heine-Borel. Suppose first that A is bounded. If A is closed, then taking $K = A$ works. Otherwise, consider $\overline{A} \setminus A$. Find V_n so that $\overline{A} \setminus A \subseteq V_n$ and $\mu(V_n) \leq \mu(\overline{A} \setminus A) + 1/n$. Let $K = \overline{A} \setminus V_n$. This is a compact set. Note that $K \subseteq A$. Then $\overline{A} = K \cup V_n$ and so $\mu(\overline{A}) = \mu(K) + \mu(V_n) \leq \mu(K) + \mu(\overline{A} \setminus A) + 1/n$. Intersecting with A, $\mu(A) \geq \mu(K) = \mu(A) - \mu(A \cap V_n)$. Note $\mu(V_n) = \mu(V_n \cap A) + \mu(V_n \setminus A)$. Rearraning, $\mu(A \cap V_n) = \mu(V_n) - \mu(V_n \setminus A) \leq \mu(V_n) - \mu(\overline{A} \setminus A)$.[31] Substituting this into our previous equation: $\mu(A) \geq \mu(K) = \mu(A) - \mu(A \cap V_n) \geq \mu(A) - \mu(V_n) + \mu(\overline{A} \setminus A) \geq \mu(A) - 1/n$. And we are done with the bounded case. If A is unbounded but $\mu(A)$ is finite. Let $A_k = A \cap [-k, k]$ and

[30] I don't understand what was going on, so if someone could shed some light please DM me.

[31] "Is it going to work? Let's see. *checks Folland* Ah. No. *starts over*" - Nabu

proceed as in a similar manner as the proof of outer regularity. If $\mu(A) = \infty$, we need to finite compact sets $K \subseteq A$ which have arbitrarily large measure. But this is clear after passing to the $A_k \cap [-k, k]$ again.

Similarly to the case of outer regualrity, we can ask whether we can always write $A = F \cup N$[32] where F is F_σ and N has measure zero. Pick compact $K_n \subseteq A$ so that $\mu(A \setminus K_n) < 1/n$. Put $F = \bigcup_{n=1}^{\infty} K_n$. In the finite measure case, $\mu(A \setminus F) = 0$, and we are done. If $\mu(A) = \infty$ then pass to $A_j = A \cap [j, j+1]$.

To summarize:

Theorem 32. *Let A be measurable. Then:*

1. *There exists a G_δ set G so that $A \subseteq G$ and $\mu(G \setminus A) = 0$.*
2. *There exists a F_σ set F such that $\mu(A \setminus F) = 0$.*

Which class is bigger: the Borel sets or the measure zero sets? The latter. Open subsets of $\mathbb{R}$ are countable unions of open intervals and so have cardinality capped by that of the continum. However, there are at least as many null sets as subsets of the Cantor set, which is larger than that of the continuum. He then talked about how to construct the Cantor set, show it has the cardinality of the continuum, show that it has measure zero. A counting argument then concludes the aforementioned remark. There are a great many constructions of unintuitive counterexamples from the Cantor set and its generalizations, we omit these due to a combination of: (1) assuming the reader has seen these before and (2) that they are in Folland.[33]

Let us now move on to measurable functions. Let $\mathbb{R}$ have the Borel σ-algebra, $\mathcal{M}$, and the Borel measure, μ. Let $f : \mathbb{R} \to \mathbb{R}$. In Lebesgue theory, f is said to be measurable if $f^{-1}((-\infty, \alpha))$ is measurable for all $\alpha \in \mathbb{R}$. If this seems like an arbitrary choice, that's because it is!

Theorem 33. *The following are equivalent:*

1. *For all α, $f^{-1}((-\infty, \alpha))$ is measurable.*
2. *For all α, $f^{-1}((-\infty, \alpha])$ is measurable.*
3. *For all α, $f^{-1}((\alpha, \infty))$ is measurable.*
4. *For all α, $f^{-1}([\alpha, \infty))$ is measurable.*

Moreover, any of the above imply that for all $f^{-1}(\{\alpha\})$ is measurable for all α.

Proof. Any of the conditions imply that the preimage under f of a Borel set is measurable, since unions, intersections, etc. pass through preimages and any of the collections above generate the entire Borel σ-algebra, by **Proposition 4**. □

As a corollary to **Theorem 33**, we have that any continuous function is Lebesgue measurable. The converse is false: consider the characteristic function of $[0, 1]$.

[32] Don't be misled by this notation, this lecture was not fun.

[33] This is not to say that the rest of these notes, or indeed, Nabu's lectures in general, cannot be found in Folland.

7 Day 7 - Measurable functions

Last time: measurable functions $f : \mathbb{R} \to \overline{\mathbb{R}}$.

Suppose we have measurable spaces $(X, \mathcal{M})$ and $(Y, \mathcal{N})$. We say that $f : X \to Y$ is $\mathcal{M} - \mathcal{N}$-measurable if $f^{-1}(W) \in \mathcal{M}$ for all $W \in \mathcal{N}$. Our definition from last time is then translated as $f : (\mathbb{R}, \mathcal{L}(\mathbb{R})) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ being $\mathcal{L}(\mathbb{R}) - \mathcal{B}(\mathbb{R})$-measurable. Is this the same as being $\mathcal{L}(\mathbb{R}) - \mathcal{L}(\mathbb{R})$-measurable? The only difference is the null sets, which seems innocent enough. If f is continuous then f is $\mathcal{L}(\mathbb{R}) - \mathcal{B}(\mathbb{R})$-measurable. Is is true that, assuming f is continuous, that $f^{-1}(N)$ should be measurable for N a null set? No. "Consult our major source of counterexamples." Let $\mathcal{C}$ be the ternary Cantor set and define $Y : \mathcal{C} \to [0, 1]$ by taking $x = \sum_{i=1}^{\infty} a_i 3^{-i}$ to $\sum_{i=1}^{\infty} a_i 2^{-1} 3^{-i}$. We can extend this to a function $f : [0, 1] \to [0, 1]$ by making it constant on the deleted intervals. "The Devil's staircase" Let $g(x) = f(x) + x$. g is strictly increasing and continuous and so invertible and $h = g^{-1}$ is also continuous. Clearly $\mu(g([0, 1] \setminus \mathcal{C})) = 1$. But also $\mu(g(\mathcal{C})) = 1$, and so $g(\mathcal{C})$ possesses a non-measurable subset, say T. $N = g^{-1}(T) = h(T) \subset \mathcal{C}$ and $\mu(N) = 0$. All together, $h^{-1}(N) = T$ which is not measurable even though h is continuous.

Consider now $f : (\mathbb{R}, \mathcal{B}(\mathbb{R})) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Being measurable then means the preimage under f of a Borel set is a Borel set; we call this being Borel measurable. Is this meaningfully different from begin Lebesgue measurable? Yes. All continuous functions are Borel measurable. The aforementioned example of $T = h^{-1}(N)$ then would imply that if N were Borel then T would be measurable, which is not the case.

Let's see another counterexample. Let $f, g : \mathbb{R} \to \mathbb{R}$ with g measurable and f continuous. Then $F(x) = f(g(x))$ is measurable, but the T and N from before witness that $H(x) = g(f(x))$ is not.

Let $(X_1, \mathcal{M}_1), (X_2, \mathcal{M}_2)$ be measurable spaces. Consider the product $(X_1 \times X_2, \mathcal{M}_1 \otimes \mathcal{M}_2)$. Let $f : (Y, \mathcal{N}) \to (X_1 \times X_2, \mathcal{M}_1 \otimes \mathcal{M}_2)$. Similarly to the case of the product topology,

Proposition 34. *The projections $\pi_i : (X_1 \times X_2, \mathcal{M}_1 \otimes \mathcal{M}_2)$ are measurable. Moreover, f is measurable if and only if the $f \circ \pi_i$ are measurable.*

Proof. The first claim and if of the second claim essentially follow by the definition of $\mathcal{M}_1 \otimes \mathcal{M}_2$. The only if of the second claim boils down to the first claim and the fact that the composition of measurable functions is measurable. □

Proposition 35. *Let $f, g : \mathbb{R} \to \mathbb{R}$ be measurable. Then $f + c, cf, f + g, fg, \max\{f, g\}$, $\min\{f, g\}$ are all also measurable.*

Proof. The first two claims are clear. To show that $f+g$ is measurable, notice that $f(x)+g(x) < a$ if and only if there is a rational $r \in \mathbb{Q}$ so that $f(x) < r$ and $g(x) < a - r$. "Let's think about it. *immediately checks Folland" And so we may write $(f + g)^{-1}((-\infty, a))$ as the union over $r \in \mathbb{Q}$ of the intersection of $f^{-1}((-\infty, r))$ and $g^{-1}((-\infty, a - r))$. And we are done. For fg, we write $fg = ((f+g)^2 - (f-g)^2)/4$ and thus reducing to showing that the square of a measurable function is measurable, which is clear. The max case corresponds to taking an intersection and the min case corresponds to taking a union. These can both be generalized to supremums (and infimums) over countable families. □

What about limits? Write $\lim_{n\to\infty} f_n(x) = \sup_{k\in\mathbb{N}} \inf_{n\geq k} f_n(x)$ and conclude from the previous case. There are only "minor technical difficulties" when handling the preimage of infinity.

Definition 36. *A property of a function is said to be "almost everywhere" if the set of x so that $f(x)$ does not possess the property has measure zero. We will often abbreviate this as a.e.*

Proposition 37. *If f is measurable and $f = g$ a.e. (say $N = \{x \ : \ f(x) \neq g(x)\}$) then g is measurable.*

Proof. To see this, $\{x \ : \ g(x) < a\} = \{x \ : \ f(x) < a\} \cup \{x \in N \ : \ g(x) < a\}$, and so if the measure is complete we're in business. □

A (random) aside: The concept of measurability can be easily extended to $\mathbb{C}$ since it is "basically two copies of $\mathbb{R}$." The results from above remain true.

"Now I want to tell you about three Littlewood principles."[34]

1. "Every measurable set of finite measure[35] is nearly the finite union of intervals."
2. "Every measurable function is nearly continuous."
3. "Every pointwise convergent sequence of functions is nearly uniformly convergent."

The second principle is Lusin's theorem and the third principle is Ergoroff's theorem, both of which you may recall from MAT357.

Let us give a proof of the first principle.

Proposition 38. *Fix $\varepsilon > 0$. Let E be Lebesgue measurable. Then there exists a finite union F of intervals such that $\mu(E \Delta F) < \varepsilon$.*

Proof. Pick a countable collection I_j of intervals so that $E \subseteq \bigcup_{j=1}^{\infty} I_j$ and $\sum_{j=1}^{\infty} \mu(I_j) < \mu(E) + \varepsilon/2$. Pick n so that $\sum_{j=n+1}^{\infty} \mu(I_j) < \varepsilon/2$.[36] Let $F = \bigcup_{j=1}^{n} I_j$. Then,

$$\mu(F \setminus E) \leq \mu\left(\bigcup_{j=1}^{\infty} I_j \setminus E\right) \leq \sum_{j=1}^{\infty} \mu(I_j) - \mu(E) < \varepsilon/2$$

and

$$\mu(E \setminus F) \leq \mu\left(\bigcup_{j=1}^{\infty} I_j \setminus F\right) \leq \mu\left(\bigcup_{j=n+1}^{\infty} I_j\right) \leq \sum_{j=n+1}^{\infty} \mu(I_j) < \varepsilon/2$$

And so $\mu(E \delta F) \leq \mu(E \setminus F) + \mu(F \setminus E) < \varepsilon$ and we are done. □

[34] He delivered this part like a bedtime story. "This Littlewood principle went to the market, this littlewood principle stayed home..."

[35] This assumption is necessary, consider $E = \bigcup_{n \in \mathbb{Z}} (n - 1/3, n + 1/3)$.

[36] This is where we use the assumption that $\mu(E) < \infty$.

8 Day 8 - Simple functions, integration

Last time: the three Littlewood principles.

A simple function is a map $f : X \to \overline{\mathbb{R}}$ or $[0, \infty]$ leaving a measurable space $(X, \mathcal{M})$ such that: (1) the set of values of f is finite, say $a_1, ..., a_n$, and (2) the sets $f^{-1}(a_j)$ are measurable.

Such functions may be written as $\varphi(x) = \sum_{j=1}^{N} a_j \chi_{E_j}$ where the E_j are disjoint and measurable and χ_{E_j} is the characteristic function of E_j.

We begin our theory of integration as follows. We expect $\int_X \chi_E = \mu(E)$ and that integration should be linear $\int_X a\chi_{E_1} + \chi_{E_2} = a\mu(E_1) + \mu(E_2)$. And in this way we define integration for all simple functions. For $f : X \to [0, \infty]$, the idea is to then define $\int_X f := \sup\{\int_X \varphi \ : \ 0 \leq \varphi \leq f$, with φ simple$\}$. For more general $f : X \to \overline{\mathbb{R}}$, we split f into its positive and negative parts f_+, f_- and then define $\int_X f = \int_X f_+ - \int_X f_-$.

We also want to integrate over measurable subsets. Suppose $E \subseteq X$ is measurable and φ is a simple function. What should $\int_E \varphi$ be? The natural definition is $\int_E \varphi := \int_X \varphi\chi_E$. One may check that $\varphi\chi_E$ is also a simple function. And then as before we define $\int_E f := \int_X f\chi_E$ for $f : X \to [0, \infty]$ and using its positive and negative parts for $f : X \to \overline{\mathbb{R}}$.

For this definition to be good, we need to know that we can reasonably appoximate measurable functions by simple ones.

Theorem 39. *Let $(X, \mathcal{M}, \mu)$ be a measure space. If $f : X \to [0, \infty]$ is measurable, then there exists an increasing sequence of non-negative simple functions φ_n such that $\varphi_n \leq f$ and $\varphi_n \to f$ pointwise and $\varphi_n \to f$ uniformly on each set where f is bounded.*[37]

Proof. "This theorem is pretty simple."[38] Fix $n \geq 1$. Let $I_{n,k} = [k/2^n, (k+1)/2^n)$ for $k = 1, ..., n2^n - 1$ and let $I_{n,\infty} = [n, \infty]$. The preimages $E_{n,k} = f^{-1}(I_{n,k})$ and $F_n = f^{-1}([n, \infty])$ are measurable. Then $\varphi_n : X \to [0, \infty]$ defined by $\varphi_n(x) = \sum_{k=1}^{n2^n-1} k2^{-n}\chi_{E_{n,k}}(x) + n\chi_{F_n}(x)$ are as desired. □

As usual, splitting $f : X \to \overline{\mathbb{R}}$ into its positive and negative parts and applying the above theorem results in a similar theorem for this more general case.

Can we do better with something other than simple functions? If $X = \mathbb{R}$, we have step functions, i.e. simple functions with the E_j being intervals. By the first Littlewood principle, we can approximate any simple function by a step function, and therefore we can approximate any measurable function by step functions. What about approximating by continuous functions? Starting with a step function we could re-define it to be linear by smoothing out the jumps between steps, and we could make the intervals where this discrepancy occurs as small as desired.[39] In $\mathbb{R}^n$, it is more complicated but "completely doable." What about in an arbitrary metric space? We would need a measure which is defined on Borel sets and which has some extra regularity conditions. We will see these later. The upshot is that the argument in $\mathbb{R}^n$ extends[40] in this case, and so measurable functions

[37] Nabu mentions that there is an analog of this theorem for complex-valued functions by replacing $\varphi_1 \leq \cdots \leq f$ with $|\varphi_1| \leq \cdots \leq |f|$. It ultimately boils down to splitting f into its real and imaginary parts, splitting those into their positive and negative parts, and then applying the first theorem and some bookkeeping.

[38] Say that again...

[39] This should remind you of the statement of Egoroff's theorem.

[40] Using the Tietze extension theorem!

can be approximated by continuous functions. Recall that this is similar to another Littlewood principle: Lusin's theorem.

Let's return to integration.

Proposition 40. *Let $\varphi \geq 0$ be a simple function. Then,*

1. *If $c \geq 0$ then $\int_X c\varphi = c\int_X \varphi$.*
2. *$\int_X \varphi + \psi = \int_X \varphi + \int_X \psi$.*
3. *If $\varphi \leq \psi$ then $\int_X \varphi \leq \int_X \psi$.*
4. *$E \mapsto \int_E \varphi$ is a measure.*

Proof. Omitted. □

Proposition 41. *Let $f, g : X \to [0, \infty]$ be measurable and let E_1, E_2 be measurable. We have:*

1. *If $0 \leq g \leq f$ then $0 \leq \int_X g \leq \int_X f$.*
2. *If $E_1 \subseteq E_2$ then $\int_{E_1} f \leq \int_{E_2} f$.*

Proof. (1) follows immediately after unraveling definitions and the fact that the supremum over a larger set dominates a supremum of a smaller set. (2) quickly follows from (1). □

Let L^+ be the set of all non-negative measurable functions leaving X.

Theorem 42 (Monotone Convergence theorem)**.** *Let $f : X \to [0, \infty]$ be measurable. Suppose $f_n \in L^+$ is such that $f_1 \leq f_2 \leq \cdots$ and $\lim_{j\to\infty} f_j = f$. Then, $\lim_{n\to\infty} \int_X f_n = \int_X f$.*

Proof. For all n, $f_n \leq f$ and so $\int_X f_n \leq \int_X f$. Also, $\int_X f_n$ is increasing in n and so $\lim_{n\to\infty} \int_X f_n = \sup_{n\in\mathbb{N}} \int_X f_n$ always exists and, when taken with the previous observation, is at most $\int_X f$. Let $\alpha \in (0, 1)$ and let φ be a simple function with $0 \leq \varphi \leq f$. Let $E_n = \{x \ : \ f_n(x) \geq \alpha\varphi(x)\}$. Notice $E_n \subseteq E_{n+1}$ since the f_n are increasing. We claim that $\bigcup_{n=1}^{\infty} E_n = X$. Fix $x \in X$. If $f(x) = 0$ then $\varphi(x) = 0$ and so $f_n(x) \geq \alpha\varphi(x)$ for all $\alpha \in (0, 1)$, so $x \in E_1$. If $f(x) > 0$, we may pick some N so that, $f_n(x) > \alpha f(x) \geq \alpha\varphi(x)$ and so $x \in E_N$. Notice $\int_X f_n \geq \int_{E_n} f_n \geq \alpha \int_{E_n} \varphi$ and so in the limit (here we are using that the map $E \mapsto \int_E \varphi$ is a measure and then continuity of said measure) we have $\lim_{n\to\infty} \int_X f_n \geq \alpha \int_X \varphi$. Taking $\alpha \to 1$ and letting φ vary then yields the claim. □

As a corollary, if we can find a sequence of simple functions φ_n with $0 \leq \varphi_n \leq f$ then we can compute $\int_X f$ by computing the limits of the integrals of the simple functions, which is considerably easier.

Another corollary:

Theorem 43. *Let $f_n : X \to [0, \infty]$ be measurable and put $f = \sum_n f_n$. Then $\int_X f = \sum_n \int_X f_n$.*

Proof. If $f = f_1 + f_2$ then there exists an increasing sequence of simple functions φ_i such that $\int_X f_1 = \lim_{i\to\infty} \int_X \varphi_i$. Similarly take ψ_j for f_2. Then $\varphi_i + \psi_i$ form an increasing sequence of measurable functions below $f_1 + f_2$. Applying **Theorem 42** then finishes the proof in this case. By induction we get the claim for all finite sums. Let $F_n = \sum_{i=1}^n f_n$. Then $\int_X F_n = \sum_{i=1}^n \int_X f_i$ and so applying the previous case and **Theorem 42** yields the claim. □

Proposition 44. *$f = 0$ almost everywhere if and only if $\int_X f = 0$.*

Proof. Assume $f = \varphi$ is simple. Then $\int_X \varphi = \sum_{i\,:\,a_i>0} a_i \mu(E_i)$ where $E_i = f^{-1}(a_i)$, and we are done. If $f = 0$ almost everywhere, then any non-negative simple function below f must be 0 almost everywhere and so $\int_X f = 0$ by the previous case after unravelling definitions. Now suppose $f = 0$ not almost everywhere. Let $E_n = \{x \,:\, f(x) > 1/n\}$, and put $E = \bigcup_{n=1}^{\infty} E_n$. Then E and at least one of the E_n have positive measure. Then $\int_X f \geq \int_{E_n} f \geq \mu(E_n)/n > 0$. □

As a corollary, we can modify the statement of the monotone convergence theorem:

Theorem 45. *Assume we have $f_n : X \to [0, \infty]$ an increasing sequence of measurable functions. Suppose further that $f = \lim_{n\to\infty} f_n$ almost everywhere. Then $\int_X f = \lim_{n\to\infty} \int_X f_n$.*

Proof. Let $E = \{x \,:\, f_n(x) \to f(x)\}$. $\mu(E^c) = 0$. By **Theorem 42** $f_n\chi_E \to f\chi_E$. Also, $f_n - f_n\chi_E = 0$ almost everywhere and $f - f\chi_E = 0$ almost everywhere, and so we may replace the integrals of $f_n\chi_E$ and $f\chi_E$ with f_n and f respectively per the previous proposition. □

Theorem 46 (Fatou's lemma)**.** *Suppose $f_n \in L^+$. Then $\int_X \liminf_{n\to\infty} f_n \leq \liminf_{n\to\infty} \int_X f_n$.*

Proof. Let $F_k(x) = \inf_{n\geq k} f_n(x)$. These are measurable. For all $i \geq k$, $F_k(x) \leq f_i(x)$ and so $\int_X F_k \leq \int_X f_i$ for all $i \geq k$. Therefore $\int_X \inf_{n\geq k} f_n(x) \leq \inf_{n\geq k} \int_X f_n$. Observe that the F_k are increasing and converge to $\liminf_{n\to\infty} f_n$ and so by **Theorem 42**, we have the claim after taking the limit on each side of the previous inequality. □

As a corollary, if $f_n \to f$ almost everywhere, then $\int_X f \leq \liminf_{n\to\infty} \int_X f_n$.

Next time: Dominated Convergence theorem.

9 Day 9 - Dominated Convergence theorem

Last time: more on Littlewood's three principles, integration, and integrable functions.

Recall: $\int_X f := \sup\{\int_X \varphi \ : \ 0 \leq \varphi \leq f, \ \varphi \text{ simple}\}$, where a simple function is a measurable function taking on only finitely many values, equivalently, if it is of the form $\varphi = \sum_{j=1}^n a_j \chi_{E_j}$ for measurable E_j. We also defined $\int_E f = \int_X f\chi_E$ for measurable subsets E.

We also discussed the (Lebesgue)[41] Monotone Convergence theorem and Fatou's lemma.

Notation: we shall write $\int f$ to mean $\int_X f$ when the measure space is clear from context.

Let $f : X \to \overline{\mathbb{R}}$. We may split up f into its positive and negative parts: f_+, f_- respectively. Assuming f is measurable, both f_+ and f_- are also measurable.

In this case, we write $f \in L^1$, and say f is integrable, to mean that: (1) f is measurable, (2) $\int f_+ < \infty$, and (3) $\int f_- < \infty$. In this case, we may define $\int f = \int f_+ - \int f_-$.

Theorem 47. *Let f, g be integrable. Then,*

1. $\int cf = c\int f$.
2. $\int f + g = \int f + \int g$.
3. *If E_1, E_2 are disjoint and measurable, then* $\int_{E_1} f + \int_{E_2} f = \int_{E_1 \cup E_2} f$.
4. *If $f \leq g$, then* $\int f \leq \int g$.

Proof. Essentially the same as the proof of the corresponding results for the non-negative case, and so we omit it. □

Proposition 48 (Triangle inequality for integrals)**.** *If f is integrable, then* $\left|\int f\right| \leq \int |f|$.

Proof. Write $f = f_+ - f_-$. Then,

$$\left|\int f\right| = \left|\int f_+ - \int f_-\right| \leq \int f_+ + \int f_- = \int f_+ + f_- = \int |f|$$

as claimed. □

We now move to the Dominated Convergence theorem.

Theorem 49. *Let $(X, \mathcal{M}, \mu)$ be complete. Assume $f_n \in L^1$ is a sequence of integrable functions such that: (1) $f_n \to f$ pointwise almost everywhere and (2) there exists an integrable function g such that $|f_n| \leq g$ for all n. Then, $f \in L^1$ and $\int f = \lim_{n\to\infty} \int f_n$.*[42]

Proof. First, let us say why f is measurable. If it were that $f_n \to f$ pointwise everywhere, then f, as the pointwise limit of measurable functions, would be measurable. But, $\liminf_{n\to\infty} f_n =: F$, which is also measurable by the aforementioned argument, and F and f differ on a subset of a null

[41] Nabu points out this could have two meanings: (1) it applies to the Lebesgue measure, or (2) that the result is originally due to Lebesgue. He does not really resolve this.

[42] We can strengthen this theorem to only require $|f_n| \leq g$ almost everywhere.

set, and therefore f is measurable.[43] Now, $|f_n| \leq g$ implies that $g - f_n \in L^1$ and $g + f_n \in L^1$. By Fatou's lemma,

$$\int g - \int f = \int \liminf_{n\to\infty} g - f_n \leq \liminf_{n\to\infty} \int g - f_n = \int g - \limsup_{n\to\infty} \int f_n$$

Since $g \in L^1$, we can rearange the above to obtain $\int f \geq \limsup_{n\to\infty} f_n$. Similarly, by Fatou's lemma again,

$$\int g + \int f = \int \liminf_{n\to\infty} g + f_n \leq \liminf_{n\to\infty} \int g + f_n = \int_g + \liminf_{n\to\infty} \int f_n$$

And so, again using that $g \in L^1$, we can rearrange to, all together, obtain:

$$\int f \leq \liminf_{n\to\infty} \int f_n \leq \limsup_{n\to\infty} f_n \leq \int f$$

and so we are done. □

It is not the case that $f_1 = f_2$ almost everywhere just in case $0 < \mu(\{x \ : \ f_2(x) \neq f_2(x)\}) = \mu\left(\bigcup_{n=1}^{\infty}\{x \ : \ |f_1(x) - f_2(x)| > 1/n\}\right)$. And so one of the $E_n = \{x \ : \ |f_1(x) - f_2(x)| > 1/n\}$ has positive measure. And so $\int_{E_n} |f_1 - f_2| \geq \mu(E_n)/n$. As a result, $f_1 = f_2$ almost everywhere if and only if $\int_X |f_1 - f_2| = 0$. And so the it makes sense to identify L^1 instead with equivalence classes.[44]

Theorem 50. *Let $f_n \in L^1$ be a sequence with $\sum_{n=1}^{\infty} \int |f_n| < \infty$. Then $\sum_{n=1}^{\infty} f_n$ converges to an integrable function f and $\int f = \sum_{n=1}^{\infty} \int f_n$.*

Proof. From last time, $\int \sum_{n=1}^{\infty} |f_n| = \sum_{n=1}^{\infty} \int |f_n|$. By the Monotone Convergence theorem, $\sum_{n=1}^{\infty} |f_n|$ is integrable and $\lim_{N\to\infty} \int \sum_{n=1}^{N} |f_n| = \int \lim_{N\to\infty} \sum_{n=1}^{N} |f_n|$. Consequently, we have that $g := \sum_{n=1}^{\infty} |f_n|$ is finite almost everywhere and integrable. By the Dominated Convergence theorem, we are done. □

Let us state, but not proof for now, Egoroff's theorem.

Theorem 51 (Egoroff)**.** *Let $(X, \mathcal{M}, \mu)$ be a complete measure space, that $E \subseteq X$ is measurable with $\mu(E) < \infty$, and that f_n are measurable functions such that $f_n \to f$ almost everywhere. Then, for all $\varepsilon > 0$ there exists A such that $\mu(A) < \varepsilon$ and $f_n \to f$ uniformly on $E \setminus A$.*

We will prove this next time.

[43] This argument works if the measure μ is complete. If this is not the case, we may take its completion and perform the argument, but it may not be that f is measurable in the pre-completed measure space, so there is subtlety here. Later, we will think of L^1 as clases of functions under the equivalence relation $f_1 \sim f_2 \iff f_1 = f_2$ on a subset of a set of measure zero, and then taking those classes which have an L^1 representative in the usual sense.

[44] Probably add more exposition about this later

10 Day 10 - Lusin's and Egorff's theorems

In Folland, Egoroff's theorem is formulated for $\mathbb{R}^n$ with the Lebesgue measure. But, "equally easy" we can formulate it instead using abstract measures, which is what we'll start with this lecture. Recall that Egoroff's principle is essentially that pointwise convergence is "almost" uniform convergence.

We prove the following formulation of **Theorem 51** where the measure space has finite measure.

Theorem 52 (Egoroff)**.** *Let $(X, \mathcal{M}, \mu)$ be a complete, finite measure space. Suppose $f_n : X \to \mathbb{R}$ are measurable and converge pointwise to $f : X \to \mathbb{R}$ almost everywhere. Then, for all $\varepsilon > 0$ there exists a measurable set A such that $\mu(A) < \varepsilon$ so that $f_n|_{X\setminus A}$ converges uniformly to $f|_{X\setminus A}$.*

Proof. For $k = 1, 2, ...$, let $U_{k,m} = \bigcup_{n\geq m}\{x \in X \ : \ |f_n(x) - f(x)| \geq 1/k\}$. For $x \in X \setminus U_{k,m}$, and for all $n \geq m$, $|f_n(x) - f(x)| < 1/k$. Fix k and consider $\bigcap_{m=1}^{\infty} U_{k,m}$. It is clear by construction that this intersection lies in the set of points x where $f_n(x)$ does not converge to $f(x)$. Also by construction we have that the $U_{k,m}$ are nested and decreasing. Since X has finite measure, by continuity of measure we have that $\lim_{m\to\infty} \mu(U_{k,m}) = 0$. Thus, we may pick $m(k)$ so that for all $m \geq m(k)$ we have $\mu(U_{m,k}) < \varepsilon 2^{-k}$. Let $A = \bigcup_{k=1}^{\infty} U_{k,m(k)}$. By subadditivity, $\mu(A) \leq \sum_{k=1}^{\infty} \mu(U_{k,m(k)}) < \varepsilon$. For $x \in X \setminus A$, there is some $k_0, N(k_0)$ so that $x \notin U_{k_0,N(k_0)}$ and so $x \notin U_{k_0,m(k_0)}$ and we are done after tracing through the definition of uniform convergence. □

The assumption of finite measure cannot be dropped. Consider the following example: let $f_n : \mathbb{R} \to \mathbb{R}$ be given by $f_n = \chi_{[n,n+1]}$. Note that f_n converges pointwise to 0 everywhere, but we cannot throw out a set of size $\varepsilon = 1$ to get uniform convergence.

In Egoroff's theroem, for some applications we want $X \setminus A$ to be closed. Assuming μ is outer regular we can make this work.

Proposition 53. *Given the hypotheses of Egoroff's theorem, if in addition μ is outer regular, we may assume that $X \setminus A$ is closed.*

Proof. Fix $\varepsilon > 0$. By **Theorem 52** (or **Theorem 51**) we can find a measurable set Λ such that $\mu(\Lambda) < \varepsilon/2$ and f_n converges uniformly to f on $E = X \setminus \Lambda$. By outer regularity, there is an open set $U \supseteq \Lambda$ such that $\mu(U) < \mu(\Lambda) + \varepsilon/2 = \varepsilon$. Then $X \setminus U \subseteq X \setminus \Lambda = E$ and so the f_n uniformly converge to f on $X \setminus U$. Since U is open, $X \setminus U$ is closed. □

Theorem 54 (Lusin)**.** *Let $(X, \mathcal{M}, \mu)$ be a measure space where X is also locally compact and Hausdorff, $\mu(X) < \infty$, and μ is a Radon measure.*[45][46] *Let $f : X \to \mathbb{R}$ (you can take the codomain to be any separable metric space) be measurable. Then, for all $\varepsilon > 0$ there exists measurable A so that $\mu(A) < \varepsilon$, $X \setminus A$ is closed, and $f|_{X\setminus A}$ is continuous. Put another way, there exists a continuous function $\varphi : X \to \mathbb{R}$ such that $\mu\{x \in X \ : \ \varphi(x) \neq f(x)\} < \varepsilon$.*

Proof. "Classic proof" Step 1, suppose $f = \sum_{i=1}^{N} a_i \chi_{E_i}$ is a simple function. By inner regularity, pick $C_i \subseteq E_i$ closed so that $\mu(E_i \setminus C_i) < \varepsilon 2^{-i-2}$. Then $\mu(X \setminus \bigcup_{i=1}^{N} C_i) < \varepsilon/2$. Note that the C_i are disjoint and f restricted to any particular C_i is constant. And so we're done. Step 2, suppose $\varphi_n \to f$ for φ_n a sequence of simple functions. For each n, there exists A_n so that $X \setminus A_n$ is closed and $\varphi_n|_{X\setminus A}$ is continuous and $\mu(X \setminus A_n) < \varepsilon 2^{-n-2}$. Let $A_0 = \bigcup_{n=1}^{\infty} A_n$. Then $\varphi_n|_{A_0}$ are

[45] Finite on compact sets, inner regular, outer regular, and defined on all Borel sets.

[46] He also probably wants X to be a separable metric space.

continuous. By Egoroff's theorem, we can find $B \subseteq X \setminus A_0$ so that $\mu(B) < \varepsilon/2$ and the $\varphi_n|_{X\setminus A}$ converge uniformly to $f_{X\setminus A}$. Since the uniform limit of continuous functions is continuous, we are done. □

He then outlined a "contemporary proof" which proceeds by analyzing how to make the preimages of $f^{-1}((a,b))$ open in some closed subspace C. But he ran out of time.[47]

Note, this does not imply any sort of continuity about the original function. Consider $f = \chi_{\mathbb{Q}\cap[0,1]}$. Note that f is discontinuous everywhere but the conclusion of Lusin's theorem is still satisfied.

[47] DM me if you have notes that cover what he tried.